

Project Description

“Hospital Management System”

“Hospital Management System ”

The Hospital Management System can help any hospital by enhancing productivity, lowering paperwork, and enhancing patient care. The Hospital Management System in C++ simplifies the process for hospitals to manage medical records, billing, and appointments. Among the information stored in the system are vital signs, medical history, medication details, laboratory results, and billing information. A hospital management system that keeps records of doctors, their appointments, patients, and staff.

The objective of the above idea is to design and implement a solution that simplifies the management and operations of a hospital.

Using C++ programming language, its features, and Object Oriented features to implement the idea.

Below are the minimum classes required to be implemented in the project, increasing reasonable classes are recommended to cover more OOP features.

You should implement at least the following 4 classes:

1. **User:**

A user has a user ID, username and password to verify sign-ins.

2. **Doctor:** stores information about the doctor (5 personal information at least) and a schedule time.

And 4 methods (Add a slot – remove a slot – display available slots – getDetails – updateDetails) at least. Also add needed setters and getters.

3. **Patient:** stores information about the patient (5 personal information at least) and an appointment if the patient has one. And at least 4 methods (MakeAppointment – CancelAppointment – getDetails – updateDetails) . Also add needed setters and getters.
4. **Admin:** staff can add patients, remove patients, add doctors, and remove doctors.
5. **Appointment:** stores information about Appointment (Id - Doctor – slot). And at least one method to display Appointment information.

Notes:

- Your program should have 3 views: admin, doctor, and patient.
- The patient can view appointments, add an appointment, and cancel an appointment.
- The doctor can add slots, remove slots, see available slots

General Constraints:

- All your C++ code files (source and header files) should be in one folder named "SRC".
- Just ONE of the team members must submit the project deliverables on Canvas.
- Specifications: The program meets all the requirements.
- Execution: The program runs successfully.
- Code Design: The program applies the basic concepts of OOP correctly, such as encapsulation, abstraction, inheritance, and polymorphism.
- Readability: The code is well organized and easy to follow .
- Documentation: The code is well documented and clearly explained .
- Report: The report is well-written and meets all the requirements .
- Organize your classes in different files (.h and .cpp) for better organization.
- The evaluation will be mainly based on the student's ability to use and apply OOP concepts and the explanation of the code.
- You must apply exception handling.
- Add more classes as needed, Also add more functions as needed in the existing class.

Project Deliverables:

Phase 1: Deliver the UML for the project as a PDF on Canvas. The document submitted will have a team ID as its name (example: team12.pdf).

Team IDs will be announced after registration of teams.

Phase 2: Each project should submit on Canvas a **zip file** named by the team ID (team#.zip) that includes:

1. A README.txt file, corresponding to the "user's manual".
This file should contain: first team member names and IDs, then describes the package in a few paragraphs, how to run and use the program.
2. A "SRC" folder contains the C++ code files (source files and header files)
3. A contribution.txt that has:
 - a) Team member names and IDs
 - b) Workload distribution within the team (the role of each member)

Teams:

Work in groups of 4-5 students (mixing between groups is allowed).

Plagiarism:

Plagiarism is a serious academic offense and students who share code with others or use any source code from the internet, will lose the project grade.

Bonus:

- using files
- Doctors have specialties and patients can search for available doctors in a certain specialty and then their available slots, or search for a slot first, then find doctors in a certain specialty.

Best of luck!